Model-based RL

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CS 1840 Fall 2026

Announcements

HW3 will be released on Friday!

Midterm will be in 2 weeks on Mon 10/26!

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Finishing Up From Last Lecture



1. Q Learning

2. Revisit TD: Off-policy TD Learning

TD Learning

Given (s, a, r, s'), where $a \sim \pi(\cdot \mid s), s' \sim P(\cdot \mid s, a)$, TD updates:

$$\hat{V}^{\pi}(s) \Leftarrow \hat{V}^{\pi}(s) + \eta \left(r + \gamma \hat{V}^{\pi}(s') - \hat{V}^{\pi}(s') \right)$$

On-policy: data is generated from the policy π itself

Off-policy: data is generated from policy π_b where $\pi_b \neq \pi$

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Q: is Q-learning off-policy or on-policy?

Motivation for off-policy evaluation

Counterfactual: what would happen if I did something different?

Off-policy TD Learning

Setting: data is generated by π_b, but we want to estimate V^π for some $\pi \neq \pi_b$

Key trick: importance weighting

$$V^\pi(s) = \mathbb{E}_{a\sim\pi(\cdot|s)}\left(r(s,a) + \gamma \mathbb{E}_{s'\sim P(\cdot|s,a)} V^\pi(s')\right)$$

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$$= \mathbb{E}_{a\sim\pi_b(\cdot|s)} \frac{\pi(a\,|\,s)}{\pi_b(a\,|\,s)}\left(r(s,a) + \gamma \mathbb{E}_{s'\sim P(\cdot|s,a)} V^\pi(s')\right)$$

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Now action is sampled from π_b

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Off-policy TD Learning

Given $(s, a, r, s'),$ where $a \sim \pi_b(\cdot \mid s), s' \sim P(\cdot \mid s, a),$

Off-policy TD updates as follows:

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Case 2: $\pi(a \mid s)$ is small but $\pi_b(a \mid s)$ is large

Recap: Planning algorithm for computing

We assumed that $P(s' \mid s, a), r(s, a) \forall s, a, s'$ are **known**

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Value Iteration:

$$Q^{t+1}(s, a) \Leftarrow r(s, a) + \mathbb{E}_{s' \sim P(\cdot|s,a)} \max_{a'} Q^{t}(s', a'), \forall s, a$$

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Policy Iteration:

$$\pi^{t+1}(s) = \arg\max_{a} Q^{\pi^t}(s, a), \text{ for all } s$$

Recap: Value-based Learning

When $P(s' \mid s, a)$ is unknown, Q-learning aims to learn $Q^\star$ directly

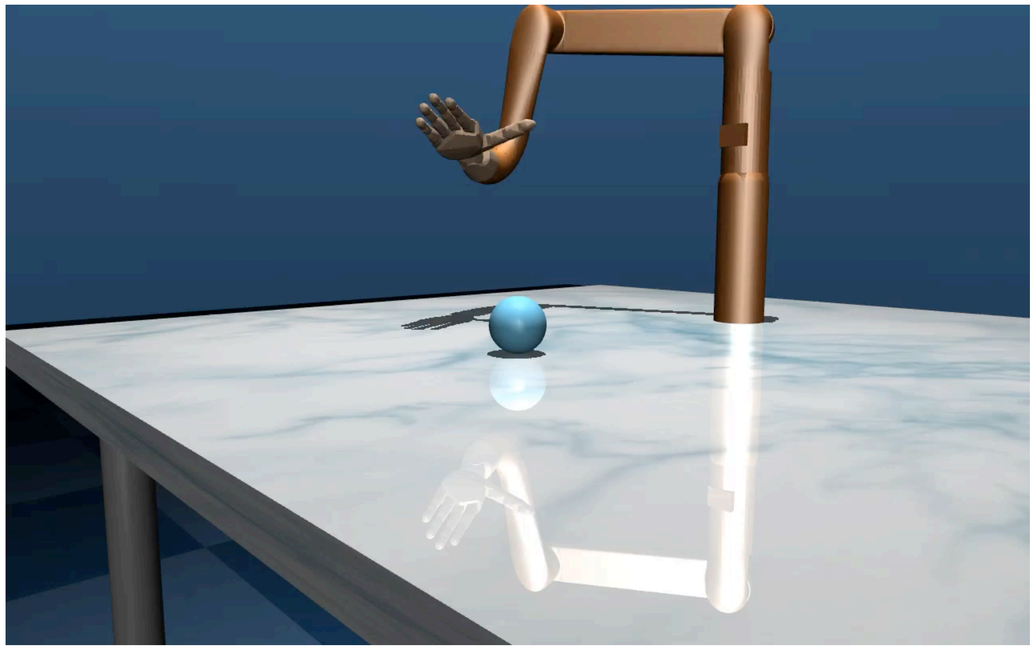
$$\hat{Q}(s,a) \Leftarrow \hat{Q}(s,a) + \left(r(s,a) + \max_{a'} \hat{Q}(s',a') - \hat{Q}(s,a) \right)$$

where $a \sim \epsilon$-greedy$(\hat{Q})$, and $s' \sim P(\,\cdot \mid s, a)$, $r = r(s,a)$

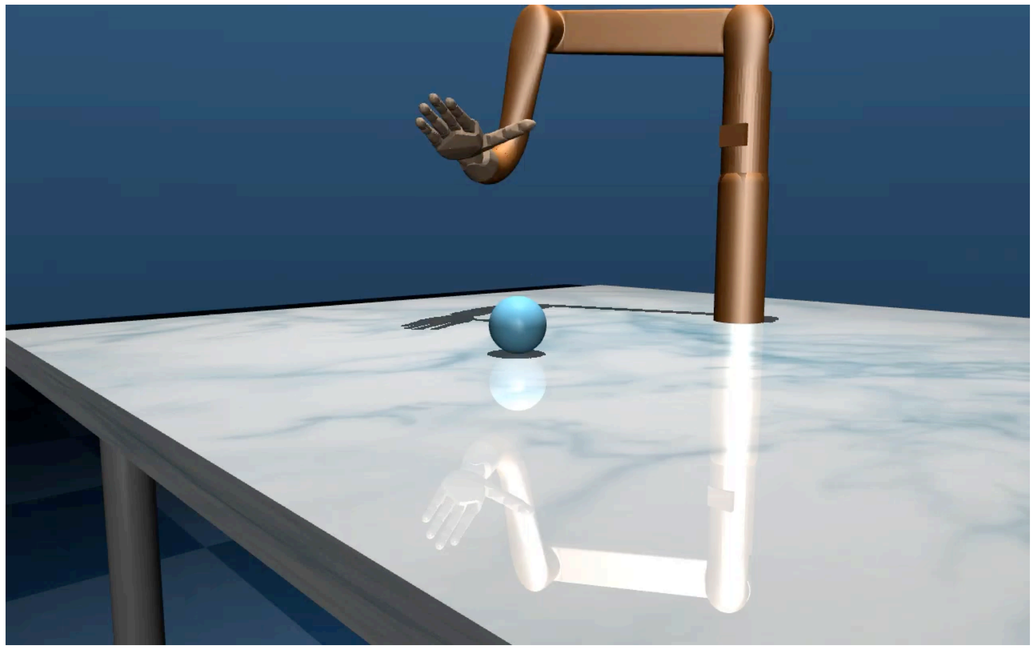
Questions for Today:

Can we **learn the transition** from data and then compute its optimal policy;
and what performance guarantee we can get?

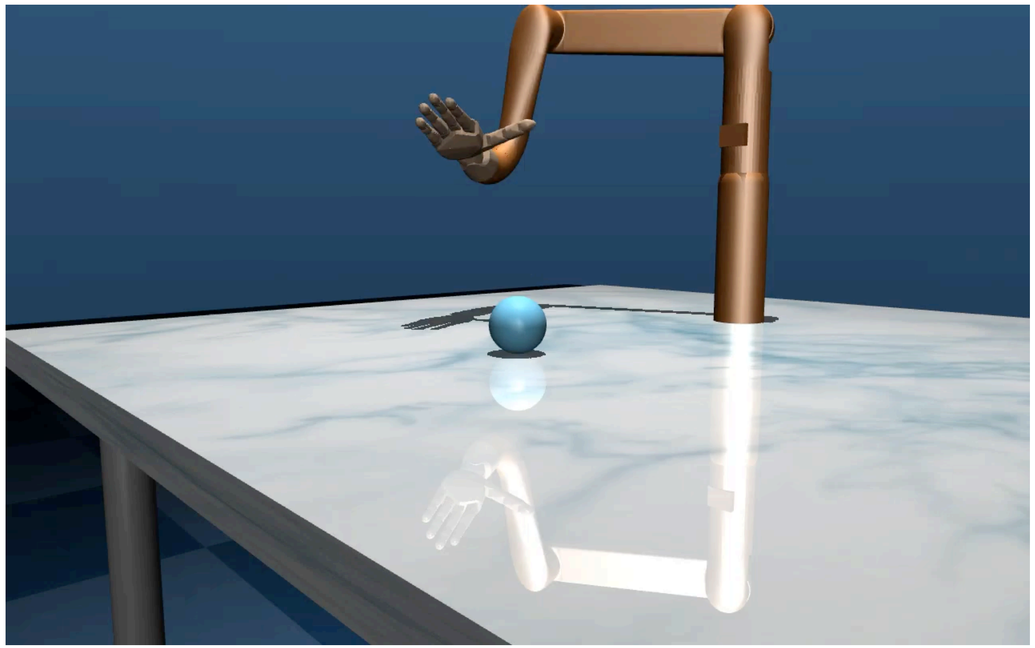
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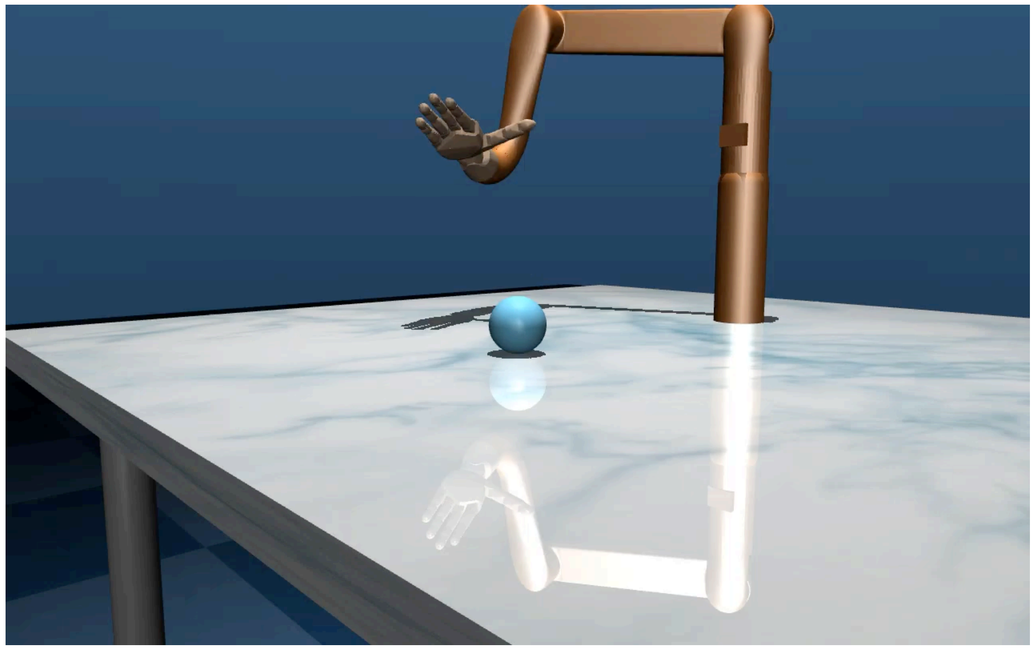
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While we cannot model the exact analytical dynamics,

we can learn it from data $\{s, a, s'\}$

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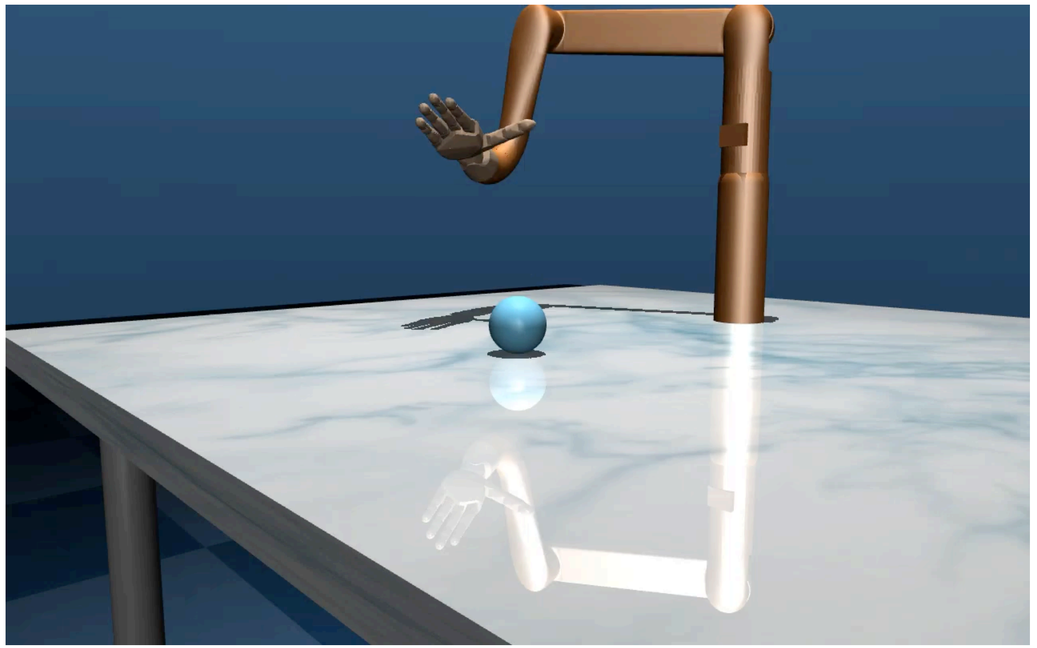
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we can learn it from data $\{s, a, s'\}$

Then we do planning: e.g.,

$$\hat{\pi}^\star = \text{VI}(\widehat{P}, r)$$

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we can learn it from data $\{s, a, s'\}$

Then we do planning: e.g.,
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(Often in practice we iterate the above process)

Motivation for Model-based Approach

Potential benefits of learning model over Q-learning

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There are cases where model is much easier to learn than value function

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Potential benefits of learning model over Q-learning

There are cases where model is much easier to learn than value function

Once model is learned, we can optimize different rewards (i.e., multi-task)

Outline:

1. Simulation lemma:
What is the performance of π under ($\widehat{P}$, r)

2. **Algorithm:** estimate $\widehat{P}$ from data
and compute $\widehat{\pi}^\star$—the optimal policy of $\widehat{P}$

3. **Analyzing** the performance $\widehat{\pi}^\star$ under (P, r)

State-action distribution

Given a policy π and s_0, we denote $\mathbb{P}_h^\pi(s, a \mid s_0)$ as the **prob of reaching (s, a) at time h, given we start at s_0 at $h = 0$**

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Denote $d_{s_0}^\pi$ as the average state-action distribution:

$$d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^\pi(s, a \mid s_0)$$

A key fundamental question in Model-based RL:

Given two transitions $\hat{P}$ and P, how would π behave differently in $\hat{P}$ and P?

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Denote:

$$\widehat{V}^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi, \widehat{P}\right]; \quad V^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi, P\right];$$

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What is the difference between $\widehat{V}^{\pi}(s_0)$ & $V^{\pi}(s_0)$?

In other words, how does the model error propagate to values

Simulation Lemma:

$$\widehat{V}^{\pi}(s_0) - V^{\pi}(s_0) = \frac{\gamma}{1-\gamma}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{s'\sim \widehat{P}(s,a)}\widehat{V}^{\pi}(s') - \mathbb{E}_{s'\sim P(s,a)}\widehat{V}^{\pi}(s')\right]$$

$$\leq \frac{\gamma}{(1-\gamma)^2}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left\| \widehat{P}(\cdot\,|\,s,a) - P(\cdot\,|\,s,a)\right\|_1$$

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$$+\gamma\mathbb{E}_{a_0\sim\pi(\cdot|s_0),s_1\sim P(s_0,a_0)}\left[\widehat{V}^{\pi}(s_1) - V^{\pi}(s_1)\right]$$

Summary so far:

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Total model disagreement over the real traces

Outline:

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1. Model fitting:

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2. Planning w/ the learned model:

$$\widehat{\pi}^\star = \mathbf{PI}\left(\widehat{P}, r\right)$$

Detour: estimating mean of Bernoulli distribution

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w/ probability at least $1 - \delta$ we will have $|\hat{p} - p| \leq O\left(\sqrt{\frac{\ln(1/\delta)}{N}}\right)$

(proof out of scope)

Model-based RL

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We can show:

$$\|\hat{P}(\cdot \mid s, a) - P(\cdot \mid s, a)\|_1 \lesssim \sqrt{\ln(1/\delta)/N}, \forall s, a$$

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$$\hat{\pi}^\star = \textbf{PI}\left(\widehat{P}, r\right)$$

Summary so far:

By collecting data (next state) at very (s, a), we build an estimator $\hat{P}$ that is close to P

(e.g., possible to show error shrinks $\approx 1/\sqrt{N}$)

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Performance of the learned policy

Now assume that $\hat{P}$ is nearly accurate, i.e., $\|\hat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1 \leq \epsilon, \forall s, a$

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(Simulation lemma)

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(Simulation lemma)

$$\leq \frac{\gamma}{(1-\gamma)^2}\left[\mathbb{E}_{s,a\sim d_{s_0}^{\pi^\star}}\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1 + \mathbb{E}_{s,a\sim d_{s_0}^{\hat{\pi}^\star}}\|\widehat{P}(\cdot|s,a) - P(\cdot|s,a)\|_1\right]$$

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$$\le \frac{2\gamma}{(1-\gamma)^2}\cdot\epsilon,$$

Summary for Today:

1. A model-based RL **Algorithm for small-size MDP**

$$\widehat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbf{1}\{s_i' = s'\}}{N}, \forall s, a; \quad \hat{\pi}^\star = \textbf{PI}\left(\widehat{P}, r\right)$$

Summary for Today:

1. A model-based RL **Algorithm for small-size MDP**

$$\widehat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbf{1}\{s_i' = s'\}}{N}, \forall s, a; \quad \hat{\pi}^\star = \mathbf{PI}\left(\widehat{P}, r\right)$$

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2. **Simulation lemma** allows us to link model error to policy's performance

3. Good model leads to a good policy, up to some error amplification based on effective horizon

Feedback and Attendance

https://bit.ly/4gYn2nC

